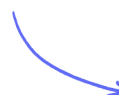


Vectors

Introduction to Column Vectors / Representing Vectors as Diagrams /
Magnitude of a Vector

Easy (6 questions)	/11
Medium (7 questions)	/19
Hard (3 questions)	/13
Total Marks	/43

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Easy Questions

1 A is the point $(1, 2)$ and $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

Find the coordinates of the point B .

(2 marks)

2 Work out $2\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} -3 \\ 5 \end{pmatrix}$.

$$\begin{pmatrix} \\ \end{pmatrix}$$

(2 marks)

3 Ahmed finds the magnitude of the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$.

From this list, select the correct calculation.

$$\sqrt{2^2 + -3^2} \quad 2^2 - 3^2 \quad \sqrt{2^2 - 3^2} \quad 2^2 + (-3^2) \quad \sqrt{2^2 + (-3)^2}$$

(1 mark)

4 $\mathbf{m} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$

Find $3\mathbf{m}$.

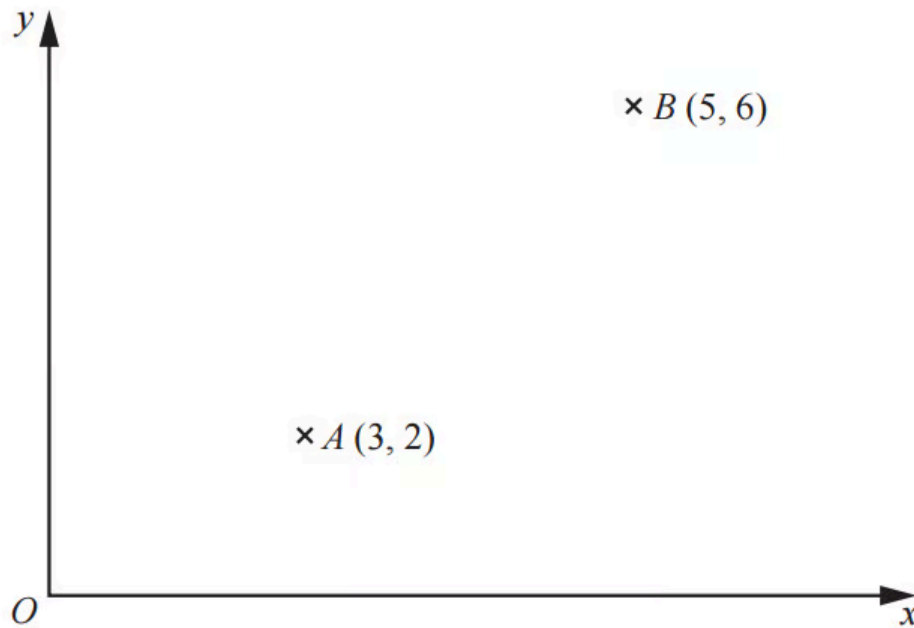
$$\begin{pmatrix} \\ \end{pmatrix}$$

(1 mark)

5 Find the magnitude of the vector $\begin{pmatrix} -1 \\ 7 \end{pmatrix}$.

(2 marks)

6



i) Find the column vector $\overrightarrow{AB}$.

$$\overrightarrow{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix} [1]$$

ii) Find $|\overrightarrow{AB}|$.

$$|\overrightarrow{AB}| = \dots\dots\dots [2]$$

(3 marks)

Medium Questions

- 1 Find the magnitude of the vector $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$.

(2 marks)

2 $\overrightarrow{VW} = \begin{pmatrix} 10 \\ -24 \end{pmatrix}$
Find $|\overrightarrow{VW}|$.

(2 marks)

3 (a) $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ $\mathbf{v} = \begin{pmatrix} -12 \\ 5 \end{pmatrix}$

Find $\mathbf{u} - 2\mathbf{v}$.

(2 marks)

(b) Find $|\mathbf{v}|$.

(2 marks)

4 $\mathbf{p} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 7 \end{pmatrix}$

i) Find $2\mathbf{p} + \mathbf{q}$.

$$\begin{pmatrix} \\ \end{pmatrix} [2]$$

ii) Find $|\mathbf{p}|$.

[2]

(4 marks)

$$5 \quad \mathbf{m} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

Find $3\mathbf{m}$.

$$\begin{pmatrix} \\ \end{pmatrix}$$

(1 mark)

$$6 \quad \mathbf{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} 14 \\ 9 \end{pmatrix}$$

i) Find $3\mathbf{a} - 2\mathbf{b}$.

$$\begin{pmatrix} \\ \end{pmatrix} [2]$$

ii) Find $|\mathbf{a}|$.

[2]

(4 marks)

7 A is the point $(-2, 1)$ and $\overrightarrow{AB} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$.

Find the coordinates of point B .

(2 marks)

Hard Questions

$$1 \quad \mathbf{p} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad \mathbf{q} = \begin{pmatrix} -5 \\ 1 \end{pmatrix}$$

(i) Work out $\mathbf{p} + 2\mathbf{q}$.

[2]

(ii) A is the point $(2, 6)$ and B is the image of point A after a translation by the vector $\mathbf{p}$.

Find the coordinates of B .

(..... ,) [1]

(iii) Find the magnitude of $\mathbf{q}$.

[2]

(5 marks)

$$2 \quad \mathbf{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} 14 \\ 9 \end{pmatrix}$$

$$m\mathbf{a} + n\mathbf{b} = \mathbf{c}$$

Write down two simultaneous equations and solve them to find the value of m and the value of n . Show all your working.

$$m = \dots\dots\dots$$

$$n = \dots\dots\dots$$

(5 marks)

3 $\vec{MT} = \begin{pmatrix} 2k \\ -k \end{pmatrix}$ and $|\vec{MT}| = \sqrt{180}$.

Find the positive value of k .

$k = \dots\dots\dots$

(3 marks)